

FCFS Disk Scheduling

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int fcfs_arr[] = {82, 170, 43, 140, 24, 16, 190};
```

```
    int total_requests = sizeof(fcfs_arr) / sizeof(int);
```

```
    int total_movements = 0;
```

```
    int current_pos = 50;
```

```
    for(int i = 0; i < total_requests; i++)
```

```
    {
```

```
        total_movements += abs(current_pos - fcfs_arr[i]);
```

```
        current_pos = fcfs_arr[i];
```

```
    }
```

```
    printf("Total arm movements (total seek distance): %d\n",  
total_movements);
```

```
    return 0;
```

```
}
```

Output:

```
G:\OS\lab6>fcfs.exe
```

```
Total arm movements (total seek distance): 642
```

SSTF Disk Scheduling

```
#include<stdio.h>

#include<stdlib.h>

int main()
{
    int sstf_arr[] = {82, 170, 43, 140, 24, 16, 190};
    int n = sizeof(sstf_arr) / sizeof(int);
    int TotalHeadMovement = 0;
    int initial=50, count = 0;
    while (count != n)
    {
        int min = 1000, d, index;
        for (int i = 0; i < n; i++)
        {
            if (sstf_arr[i] != -1)
            {
                d = abs(sstf_arr[i] - initial);
                if (min > d)
                {
                    min = d;
                    index = i;
                }
            }
        }
        TotalHeadMovement += min;
        initial = sstf_arr[index];
        sstf_arr[index] = -1; // Marking the processed request as completed
    }
}
```

```
    count++;  
  
}  
  
printf("Total head movement is %d\n", TotalHeadMovement);  
  
return 0;  
  
}
```

Output:

```
G:\OS\lab6>sstf.exe  
Total head movement is 208
```

SCAN Disk Scheduling

```
#include <stdio.h>

#include <stdlib.h>

int main() {
    int scan_arr[] = {82, 170, 43, 140, 24, 16, 190};
    int n = sizeof(scan_arr) / sizeof(int);
    int total_head_movement = 0;
    int initial_position=50, i;

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (scan_arr[j] > scan_arr[j + 1]) {
                int temp = scan_arr[j];
                scan_arr[j] = scan_arr[j + 1];
                scan_arr[j + 1] = temp;
            }
        }
    }

    int initial_index = 0;
    for (i = 0; i < n; i++) {
        if (scan_arr[i] >= initial_position) {
            initial_index = i;
            break;
        }
    }

    for (i = initial_index; i < n - 1; i++) {
        total_head_movement += abs(scan_arr[i] - scan_arr[i + 1]);
    }

    for (i = initial_index; i > 0; i--) {
```

```
        total_head_movement += abs(scan_arr[i] - scan_arr[i - 1]);  
    }  
    printf("Total head movement is %d\n", total_head_movement);  
    return 0;  
}
```

Output:

```
[Running] cd "g:\OS\lab6\" && gcc scan-algo.c -o scan-algo &&  
"g:\OS\lab6\"scan-algo  
Total head movement is 174  
  
[Done] exited with code=0 in 1.406 seconds
```

C-SCAN Disk Scheduling

```
#include <stdio.h>

#include <stdlib.h>

int main() {
    int cscan_arr[] = {82, 170, 43, 140, 24, 16, 190};
    int n = sizeof(cscan_arr) / sizeof(int);
    int total_head_movement = 0;
    int initial_position=50, i;

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (cscan_arr[j] > cscan_arr[j + 1]) {
                int temp = cscan_arr[j];
                cscan_arr[j] = cscan_arr[j + 1];
                cscan_arr[j + 1] = temp;
            }
        }
    }

    int initial_index = 0;
    for (i = 0; i < n; i++) {
        if (cscan_arr[i] >= initial_position) {
            initial_index = i;
            break;
        }
    }

    for (i = initial_index; i < n - 1; i++) {
        total_head_movement += abs(cscan_arr[i] - cscan_arr[i + 1]);
    }
}
```

```
total_head_movement += cscan_arr[n - 1];
total_head_movement += cscan_arr[0];
for (i = 0; i < initial_index; i++) {
    total_head_movement += abs(cscan_arr[i] - cscan_arr[i + 1]);
}
printf("Total head movement is %d\n", total_head_movement);
return 0;
}
```

Output:

```
[Running] cd "g:\OS\lab6\" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile &&
"g:\OS\lab6\"tempCodeRunnerFile
Total head movement is 380

[Done] exited with code=0 in 1.322 seconds
```

LOOK Disk Scheduling

```
#include <stdio.h>

#include <stdlib.h>

int main() {
    int look_arr[] = {82, 170, 43, 140, 24, 16, 190};
    int n = sizeof(look_arr) / sizeof(int);
    int total_head_movement = 0;
    int initial_position=50, i;
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (look_arr[j] > look_arr[j + 1]) {
                int temp = look_arr[j];
                look_arr[j] = look_arr[j + 1];
                look_arr[j + 1] = temp;
            }
        }
    }
    int initial_index = 0;
    for (i = 0; i < n; i++) {
        if (look_arr[i] >= initial_position) {
            initial_index = i;
            break;
        }
    }
    for (i = initial_index; i < n - 1; i++) {
        total_head_movement += abs(look_arr[i] - look_arr[i + 1]);
    }
    for (i = initial_index; i > 0; i--) {
        total_head_movement += abs(look_arr[i] - look_arr[i - 1]);
    }
    printf("Total head movement is %d\n", total_head_movement);
}
```



```
    return 0;  
}
```

Output:

```
[Running] cd "g:\OS\lab6\" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile &&  
"g:\OS\lab6\"tempCodeRunnerFile
```

```
Total head movement is 174
```

```
[Done] exited with code=0 in 0.919 seconds
```